
Integrated energy grids Assignment

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Table 1: Contribution of each member to project tasks. Repository: [GitHub link](#)

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1 Data collection and cost assumptions

1.1 Data construction methodology

The dataset combines installed capacity, hourly generation and demand, and fuel prices to build a consistent input framework for the dispatch model. Installed capacities are aggregated by technology to define the maximum available production. Hourly time series of electricity production reconstruct demand and generation profiles, ensuring internal consistency between supply and demand. Technologies are grouped into representative categories: renewable sources (wind, solar, hydro), nuclear, and thermal technologies (coal, CCGT, CHP). For the multi-country analysis, the same methodology is extended to France, Italy and Portugal, ensuring consistent demand, renewable generation and capacity representation across regions.

1.2 Capacity factors from real generation data

Per-unit capacity factors $CF_{t,h} \in [0,1]$ for wind, solar and hydro are computed by dividing each country's hourly generation series by its baseline installed capacity (no synthetic profile is used). Spain's annual capacity factors are 0.216 (Wind), 0.161 (Solar) and 0.240 (Hydro). For the dispatchable technologies (CCGT, Coal, Nuclear) availability is set to one, so the optimiser is free to dispatch up to the installed capacity. The full annual hourly profile for Spain is shown in Appendix Fig. A.2.

1.3 Technology costs

Table 2: Technology cost database used in this study. Source: DTU course 46770 Integrated Energy Grids, 2026 edition (Discount rate for Annualized costs of 5%)[1] (Techs_&_storage_cost_FOM_&_Fuel_cost.csv).

Technology	Category	Overnight (€/kW)	FOM (%/a)	Lifetime (a)	Eff. (–)	Fuel (€/MWh)	Ann. cost (€/kW/a)
Wind onshore	wind	1 182	3.0	25	1.00	0.02	83.78
Wind offshore	wind	2 506	3.0	25	1.00	0.02	177.78
Solar PV rooftop	solar	725	2.0	25	1.00	0.01	51.44
Solar PV utility	solar	425	3.0	25	1.00	0.01	30.15
Pumped hydro storage	hydro	2 000	1.0	80	0.87	0.03	102.04
Hydro reservoir	hydro	2 000	1.0	80	0.90	0.03	102.04
Run-of-river	hydro	3 000	2.0	80	0.90	0.03	153.05
CHP central	chp	600	3.0	25	0.80	87.50	42.57
CCGT	fossil	400	4.0	30	0.39	30.00	26.02
Coal	coal	3 000	3.0	40	0.40	61.52	174.61
Nuclear	nuclear	6 000	2.5	60	0.33	8.50	317.04
Battery system	storage	450	2.0	15	0.90	0.00	43.35
Hydrogen system	storage	2 800	3.0	25	0.35	0.00	198.65
Air HP (decentral)	heat_pumps	1 050	3.0	20	3.5	10.00	121.00
Air HP (central)	heat_pumps	700	3.0	20	3.5	10.00	80.67
Ground HP (decentral)	heat_pumps	1 400	3.5	20	3.5	8.57	161.33

2 Optimal capacity expansion – baseline (Part a)

Formulation. The single-country LP minimises annualised CAPEX plus variable OPEX over $T = \{\text{Wind, Solar, Hydro, CCGT, Coal, Nuclear}\}$ and $H = 8784$ hours,

$$\min_{C_t, G_{t,h}} \sum_{t \in T} FC_t \cdot C_t + \sum_{t,h} VC_t \cdot G_{t,h} \quad \text{s.t.} \quad \sum_t G_{t,h} = D_h, \quad G_{t,h} \leq CF_{t,h} C_t \text{ (ren.)}, \quad G_{t,h} \leq C_t \text{ (disp.)},$$

with $0 \leq C_t \leq C_t^{\max}$. Renewable upper bounds are 50 GW for wind/solar (due the low density of power of these technologies and the limitation on the source locations) and the historical 17 097 MW for hydro. Total annual demand is **248.8 TWh**. The LP converges to an annual cost of **€773,34 million** and the capacity portfolio in Table 3. Wind and solar reaches its 50 GW ceiling, indicating that a higher cap would still be economic for both of them but is binding for the cheaper resource. Hydro is built at its physical maximum because the LP treats reservoir water as a free dispatchable resource. Coal is uniformly excluded, it is dominated by gas under the assumed costs. Nuclear is built at 16.3 GW, providing the bulk of the firm energy outside the windy/sunny hours. The annual mix (Fig. 3) is **89 % renewable**.

Table 3: Optimal capacities for Spain – updated case (based on cost-optimised dispatch).

Technology	Capacity (MW)	Annual gen. (GWh)	Share (%)	Realised CF
Wind	50 000	94 917	38.2	0.217
Solar	50 000	69 933	28.1	0.160
Hydro	17 097	33 943	13.6	0.227
Nuclear	16 329	49 303	19.8	0.345
CCGT	9 058	695	0.3	0.009
Coal	0	0	0.0	–
Total	142 484	248 791	100	–

Dispatch and duration curves. Fig. 1-2 shows one representative week in winter and one in summer. Wind dominates the winter week with strong night-time generation, while solar produces a clear bell-shaped daytime block in summer.

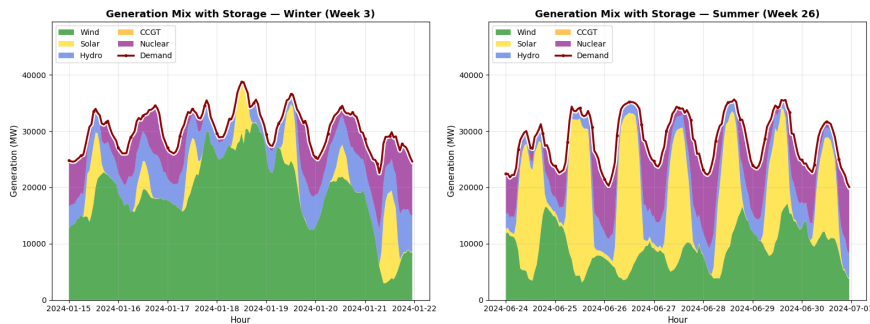


Figure 1: Optimised dispatch for one winter week (left) and one summer week (right) in Spain, baseline LP without storage.

Nuclear and CCGT cover the residual peak. The full per-technology load duration curves confirm

that nuclear runs at full power for ~ 1000 h in winter while CCGT is required for only ~ 200 h, justifying its low capacity factor. This mostly due to the high fuel cost of gas, while nuclear is really expensive on CAPEX but really profitable on its generation.

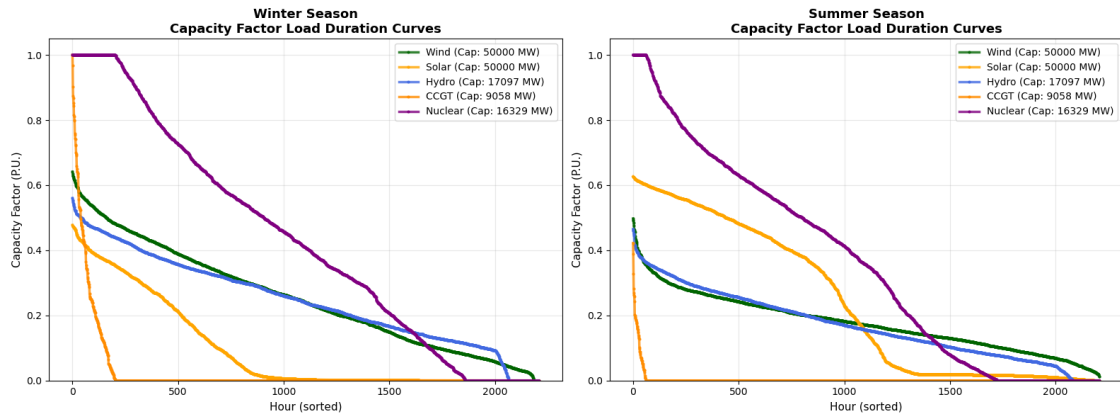


Figure 2: Technology duration curves for one winter week (left) and one summer week (right) in Spain, baseline LP without storage

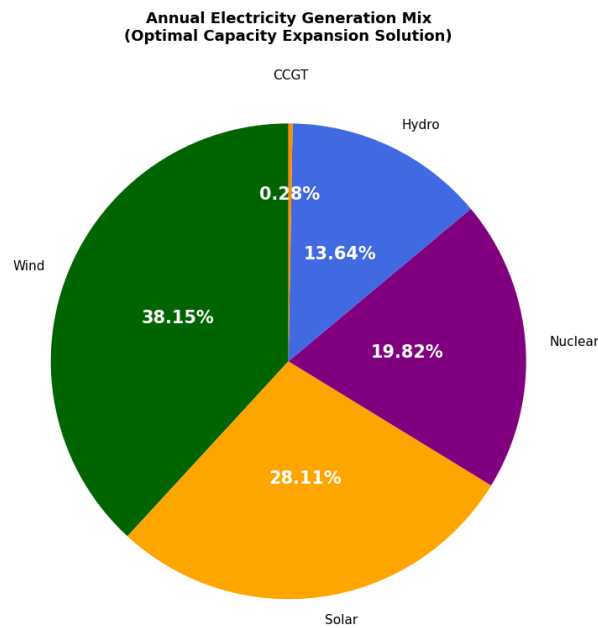


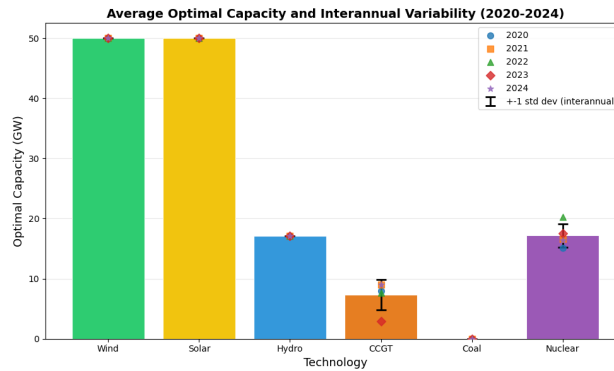
Figure 3: Annual electricity mix (baseline).

3 Interannual variability of wind and solar (Part b)

The capacity-expansion model is rerun once per weather year between 2020 and 2024, keeping the demand and cost data fixed and using the actual ENTSO-E hourly generation of that year to derive renewable capacity factors. Table 4 reports the optimal capacities; Fig. 4 shows the average and $1\text{-}\sigma$ inter-year range.

Table 4: Optimal capacities (MW) by weather year and aggregated statistics.

Year	Wind	Solar	Hydro	CCGT	Coal	Nuclear
2020	50 000	50 000	17 097	7 986	0	15 128
2021	50 000	50 000	17 097	8 968	0	16 538
2022	50 000	50 000	17 097	7 610	0	20 307
2023	50 000	50 000	17 097	2 892	0	17 516
2024	50 000	50 000	17 097	9 058	0	16 329
Mean	50 000	50 000	17 097	7 303	0	17 164
Std dev	0	0	0	2 330	0	1 818
CV	0%	0%	0%	31.9%	—	10.6%

Figure 4: Average optimal capacity per technology over the five weather years (2020–2024). Deviation bars show $\pm 1\sigma$ across years; markers are individual yearly results.

Wind, solar and hydro sit at their upper bounds in all years due to their low costs, making the capacity limits binding rather than economic. The most variable technology is **CCGT** ($\sigma/\mu \approx 32\%$), reflecting interannual differences in renewable availability: in high-yield years (e.g. 2023), strong solar output reduces gas capacity needs to ~ 3 GW, while in weaker years (2020–2021) around 9 GW is required for adequacy. Nuclear absorbs part of this variability ($\sigma/\mu \approx 10\%$) as the next-cheapest firm option once renewables are fully deployed.

The main takeaway is that renewables and nuclear cover the bulk of demand, while gas acts as a residual balancing technology, only required in extreme low-renewable conditions. Because these rare events vary significantly across weather years, CCGT capacity exhibits large fluctuations. This implies that single-year studies can mis-estimate firm capacity needs by several GW, motivating multi-year analysis for robust planning.

4 Battery and hydrogen storage integration (Part c)

Two storage technologies are now co-optimised together with the generators: a 4-hour Li-ion battery (€867/MW/yr annualised, round-trip $\sim 90\%$) and a 168-hour hydrogen system (€5 960/MW/yr, round-trip $\sim 35\%$). Both have a 25 % initial state of charge and are driven by the standard PyPSA

storage equations

$$E_h^s = E_{h-1}^s + \eta_{\text{ch}}^s Ch_h^s - \frac{1}{\eta_{\text{dis}}^s} Disch_h^s, \quad s \in \{\text{bat}, H_2\}. \quad (1)$$

Table 5: Spanish optimal capacities and total cost: baseline vs. baseline + battery + H₂.

Technology	Baseline (MW)	With storage (MW)	Δ (MW)
Wind	50 000	50 000	0
Solar	50 000	50 000	0
Hydro	17 097	17 097	0
CCGT	9 058	5 521	−3 537
Nuclear	16 329	13 120	−3 209
Coal	0	0	0
Battery (power)	—	14 108	—
Hydrogen (power)	—	330	—
Annual cost (M€)	773,34	739,82	33,52

Optimal storage capacities and impact. Adding storage modifies primarily the *firm capacity mix* rather than renewable deployment (Table 5). Wind, solar and hydro remain at their upper bounds (50 GW, 50 GW and 17 GW, respectively), while **CCGT capacity decreases from 9.1 to 5.5 GW** and **nuclear from 16.3 to 13.1 GW**. This reflects the role of storage in providing flexibility and part of the firm capacity previously supplied by thermal technologies. The optimiser installs **14.1 GW of battery** and a much smaller **0.33 GW of hydrogen**, indicating that short-term flexibility is significantly more valuable than long-duration storage in this setup.

Operational behaviour. The dispatch in Fig. 5 highlights the different timescales of operation. In a typical *summer* week, the battery cycles daily, charging during periods of high solar availability and discharging during evening peaks. The hydrogen system remains largely inactive due to the absence of prolonged stress periods. In *winter*, the battery continues to provide short-term balancing, while hydrogen contributes marginally during extended low-renewable periods. The state-of-charge profiles (Fig. 6) confirm this behaviour: the battery exhibits frequent cycling, whereas the hydrogen system operates at a much lower utilisation level.

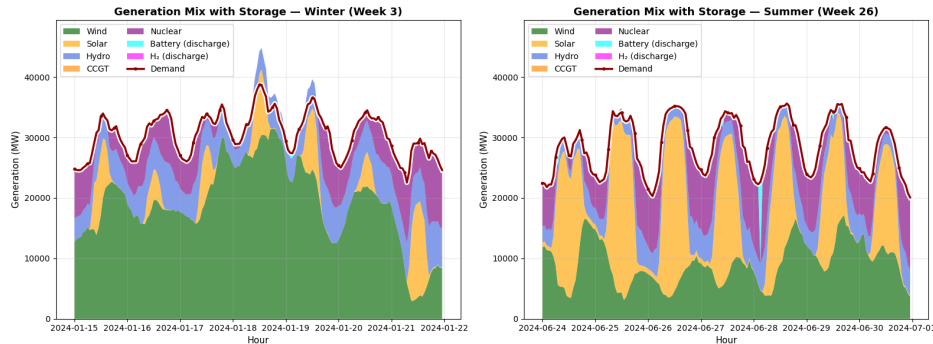


Figure 5: Optimised dispatch with battery and hydrogen storage for a representative winter (left) and summer (right) week in Spain. Cyan = battery discharge; pink = H₂ discharge.

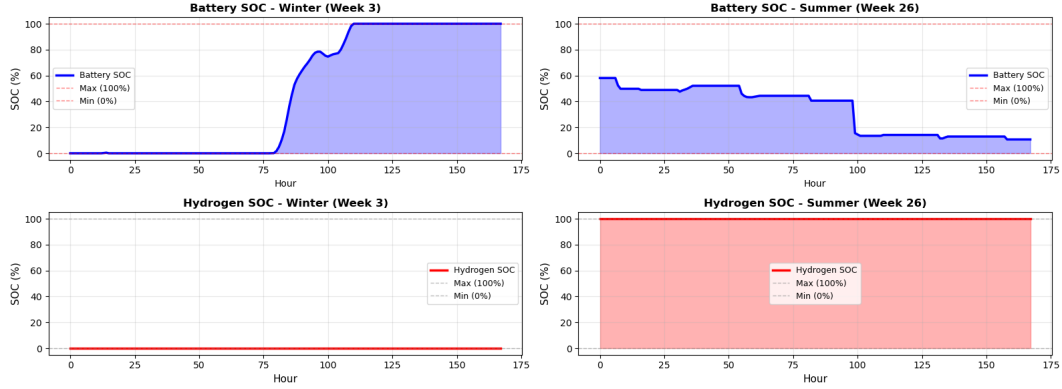


Figure 6: State-of-charge (SOC) profiles of storage systems.

Multi-timescale balancing. The system exhibits a clear hierarchy of flexibility options. Hydro reservoirs provide mid-merit balancing, while the battery handles most of the short-term variability driven by solar, cycling on a daily basis. The hydrogen system plays only a marginal role, indicating that long-duration storage is not strongly valued under the given conditions. As a result, storage primarily substitutes part of the firm capacity: both nuclear and CCGT are reduced, but not eliminated, since they are still required to cover extreme low-renewable periods. This confirms that, in this setup, short-term flexibility dominates system needs, with limited reliance on seasonal balancing.

5 Multi-country interconnection (Part d)

Spain is interconnected to France, Italy and Portugal via four 400kV HVAC links forming the closed cycle ES–FR–IT–ES (plus the radial branch ES–PT). NTC values are fixed at their 2024 values: ES–FR 3 500 MW, FR–IT 4 500 MW, ES–IT 2 000 MW, ES–PT 4 500 MW; reactance $x = 0.1$ p.u. per line. Generation capacities in all four countries are co-optimised with PyPSA’s linearised AC (DC) flow model, using country-specific demand and capacity factors and the storage technologies of Section 4. The optimum is **€4.27 billion/year** for the whole four-country system.

Multi-country formulation. The model is extended to a set of nodes $n \in \mathcal{N}$ (countries), each with its own demand time series $d_{n,t}$ and renewable capacity factors $cf_{n,k,t}$, while identical techno-economic parameters are assumed across countries. The optimisation remains a linear cost-minimisation problem, with the following additional constraints.

Nodal power balance:

$$\sum_k g_{n,k,t} + \sum_s \left(p_{n,s,t}^{\text{dis}} - p_{n,s,t}^{\text{ch}} \right) + \sum_{m \in \mathcal{N}} f_{m,n,t} - \sum_{m \in \mathcal{N}} f_{n,m,t} = d_{n,t} \quad \forall n, t$$

Transmission limits (interconnectors):

$$-F_{n,m} \leq f_{n,m,t} \leq F_{n,m} \quad \forall (n, m), t$$

Renewable generation limits:

$$0 \leq g_{n,k,t} \leq \text{cf}_{n,k,t} \cdot G_{n,k} \quad \forall n, k, t$$

where $g_{n,k,t}$ is generation of technology k at node n , $G_{n,k}$ is installed capacity, $f_{n,m,t}$ is the power flow from node n to m , and $F_{n,m}$ is the transmission capacity. These constraints enable cross-border trade and capture the spatial coupling of the system while preserving country-specific demand and resource availability.

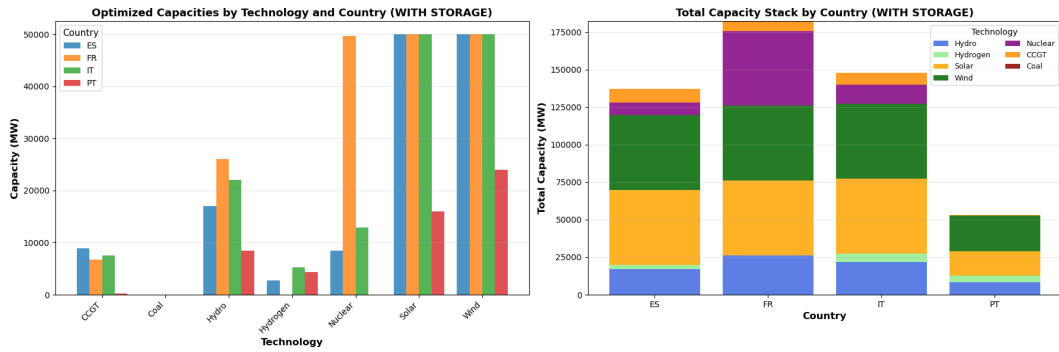


Figure 7: Optimised installed capacities (MW) with storage by technology and country

Discussion. The optimum is again deeply renewable ($\approx 71\%$ wind+solar+hydro across the four countries), with France's mix dominated by 50 GW of nuclear that exports steady baseload over the meshed network, this is due to its high demand and the limit on renewables set. The interconnectors are heavily loaded: the ES–FR line saturates during the daily Iberian solar peak and the ES–PT line stays close to its ± 4.5 GW limit for most hours (Appendix Fig. A.10). Italy is the only net importer; Spain and Portugal export their excess solar and wind respectively. Hydrogen storage is built in ES (2.8 GW), IT (5.3 GW) and PT (4.3 GW) but *not* in France, because the nuclear-dominated French residual load is already very flat.

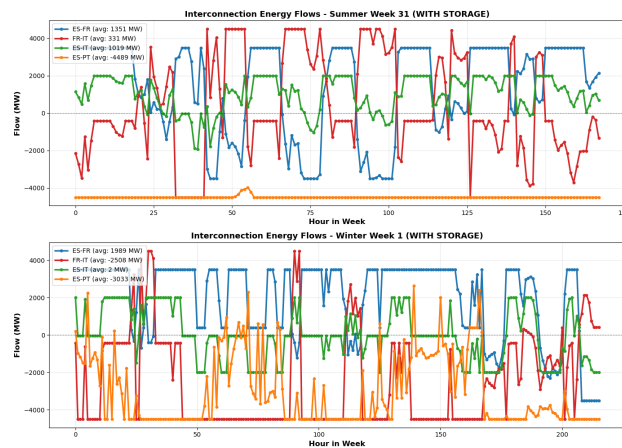


Figure 8: Interconnection energy flows between countries during summer and winter.

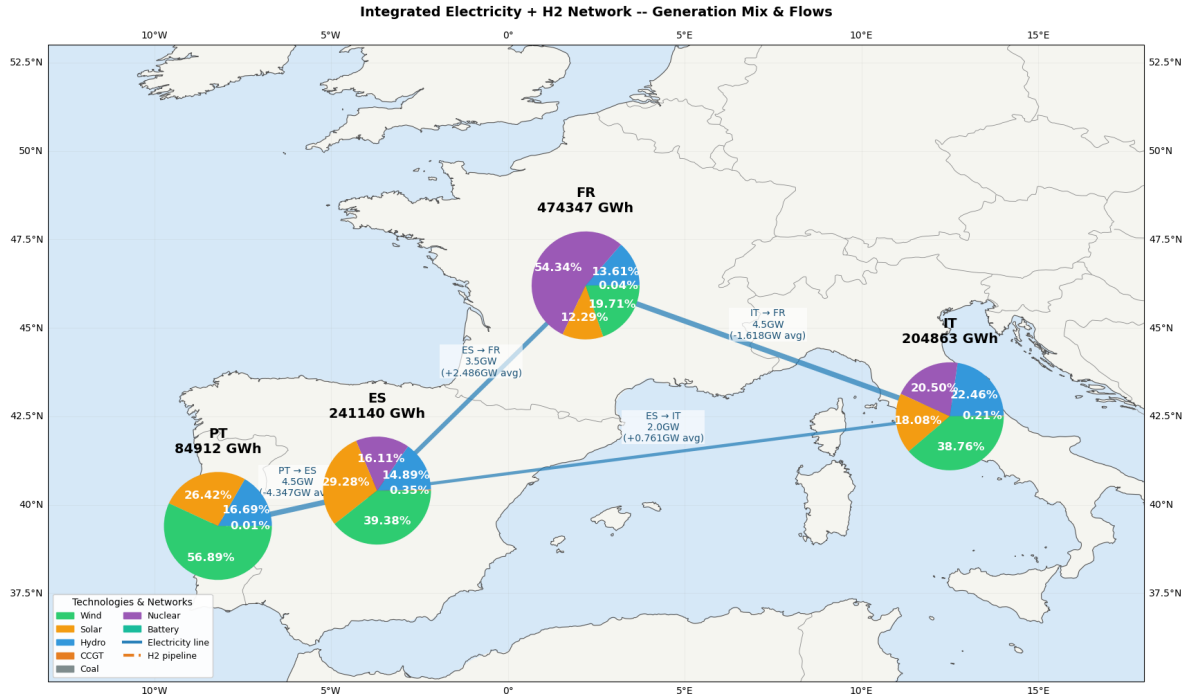


Figure 9: Average hourly electricity flows (MW) across interconnections in the multi-country system.

6 Pen-and-paper PTDF verification (Part e)

This section derives analytically the incidence matrix K , the network Laplacian L , and the Power Transfer Distribution Factor (PTDF) matrix of the four-bus system defined in the previous section. The resulting PTDF is then used to reproduce the line flows at the first simulation time step from the nodal power imbalances, and the result is compared against the PyPSA output to validate the linearised AC (DC) approximation.

6.1 Network definition

The four-bus, four-line network is shown in Fig. 10. Buses are numbered $0 = \text{ES}$, $1 = \text{FR}$, $2 = \text{IT}$, $3 = \text{PT}$. Lines are defined as:

- Line 0: $\text{ES} \rightarrow \text{FR}$ (NTC = 3500 MW, $x = 0.1$ p.u.)
- Line 1: $\text{FR} \rightarrow \text{IT}$ (NTC = 4500 MW, $x = 0.1$ p.u.)
- Line 2: $\text{IT} \rightarrow \text{ES}$ (NTC = 2000 MW, $x = 0.1$ p.u.)
- Line 3: $\text{ES} \rightarrow \text{PT}$ (NTC = 4500 MW, $x = 0.1$ p.u.)

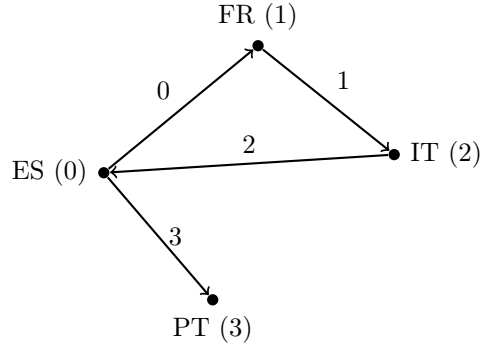


Figure 10: Reduced four-bus network used for the PTDF derivation.

All lines have equal reactance $x = 0.1$ p.u., giving a uniform susceptance $b = 1/x = 10$ p.u. Spain (bus 0) is chosen as the slack bus.

6.2 Step 1 – Incidence matrix K

The incidence matrix $K \in \mathbb{R}^{N \times L}$ (nodes $\times$ lines) encodes the network topology using the convention: $K_{n,\ell} = +1$ if line ℓ leaves node n , $K_{n,\ell} = -1$ if line ℓ enters node n , and 0 otherwise.

For our four nodes and four lines:

$$K = \begin{pmatrix} +1 & 0 & -1 & +1 \\ -1 & +1 & 0 & 0 \\ 0 & -1 & +1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{array}{l} \leftarrow \text{ES (0)} \\ \leftarrow \text{FR (1)} \\ \leftarrow \text{IT (2)} \\ \leftarrow \text{PT (3)} \end{array} \quad (2)$$

Reading each column: Line 0 (ES→FR) leaves ES (+1) and enters FR (−1); Line 1 (FR→IT) leaves FR (+1) and enters IT (−1); Line 2 (IT→ES) leaves IT (+1) and enters ES (−1); Line 3 (ES→PT) leaves ES (+1) and enters PT (−1).

6.3 Step 2 – Diagonal susceptance matrix B

Since all lines have equal susceptance $b = 10$ p.u., the diagonal matrix B is:

$$B = \text{diag}(b_0, b_1, b_2, b_3) = 10 \cdot I_4 = \begin{pmatrix} 10 & 0 & 0 & 0 \\ 0 & 10 & 0 & 0 \\ 0 & 0 & 10 & 0 \\ 0 & 0 & 0 & 10 \end{pmatrix}. \quad (3)$$

6.4 Step 3 – Full network Laplacian L

The full Laplacian (also called the nodal admittance matrix in the DC approximation) is:

$$L = K B K^T. \quad (4)$$

Computing explicitly (since $B = 10I$, this simplifies to $L = 10 K K^T$):

$$L = 10 \begin{pmatrix} 3 & -1 & -1 & -1 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 2 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}. \quad (5)$$

Each diagonal entry L_{nn} equals the sum of susceptances of all lines connected to bus n ; each off-diagonal entry L_{nm} equals minus the sum of susceptances of lines directly connecting n and m . For example, ES (bus 0) is connected to FR, IT, and PT, so $L_{00} = 10(b_0 + b_2 + b_3) = 10 \times 3 = 30$.

6.5 Step 4 – Reduced Laplacian L_R (slack bus removal)

With Spain (bus 0) as the slack bus, we remove row 0 and column 0 from L to obtain the reduced Laplacian $L_R \in \mathbb{R}^{3 \times 3}$ (over the non-slack buses FR, IT, PT):

$$L_R = 10 \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (6)$$

Note that PT (bus 3) is connected only to ES (the slack bus), so it decouples from the FR–IT sub-system, producing the block-diagonal structure.

6.6 Step 5 – Inverse of the reduced Laplacian L_R^{-1}

We invert L_R analytically. For the 2×2 block (FR–IT sub-system):

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}^{-1} = \frac{1}{4-1} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}. \quad (7)$$

The PT block is trivially $[1]^{-1} = [1]$. Combining and dividing by the factor of 10 from L_R :

$$L_R^{-1} = \frac{1}{10} \begin{pmatrix} 2/3 & 1/3 & 0 \\ 1/3 & 2/3 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (8)$$

6.7 Step 6 – Reduced incidence matrix K_R

K_R is obtained from K by removing the column (bus) corresponding to the slack bus (ES, column 0):

$$K_R = \begin{pmatrix} 0 & -1 & +1 \\ +1 & 0 & 0 \\ -1 & +1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{array}{l} \leftarrow \text{Line 0} \\ \leftarrow \text{Line 1} \\ \leftarrow \text{Line 2} \\ \leftarrow \text{Line 3} \end{array} \quad (9)$$

The columns now correspond to the non-slack buses FR (1), IT (2), PT (3).

6.8 Step 7 – PTDF matrix

The PTDF matrix is defined as:

$$\text{PTDF} = B K_R^T L_R^{-1}, \quad (10)$$

where $B K_R^T \in \mathbb{R}^{L \times N-1}$ maps bus injections to raw line flows, and L_R^{-1} converts injections into voltage angles. Since $B = 10I$ and L_R^{-1} carries a factor $1/10$, the factors cancel exactly, leaving a dimensionless result for uniform reactance networks.

Computing $K_R^T L_R^{-1}$:

$$K_R^T = \begin{pmatrix} 0 & +1 & -1 & 0 \\ -1 & 0 & +1 & 0 \\ +1 & 0 & 0 & -1 \end{pmatrix}, \quad (11)$$

$$\begin{aligned} K_R^T L_R^{-1} &= K_R^T \cdot \frac{1}{10} \begin{pmatrix} 2/3 & 1/3 & 0 \\ 1/3 & 2/3 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \frac{1}{10} \begin{pmatrix} 1/3 & -1/3 & 0 \\ -2/3 & -1/3 & 0 \\ 2/3 & 1/3 & 0 \\ 0 & 0 & -1 \end{pmatrix}. \end{aligned} \quad (12)$$

Multiplying by $B = 10I$ (the factor of 10 cancels the $1/10$):

$$\boxed{\text{PTDF} = \begin{pmatrix} -2/3 & -1/3 & 0 \\ +1/3 & -1/3 & 0 \\ +1/3 & +2/3 & 0 \\ 0 & 0 & -1 \end{pmatrix}} \quad \text{columns: FR (1), IT (2), PT (3)}. \quad (13)$$

Physical interpretation. Each entry $\text{PTDF}_{\ell,n}$ gives the fraction of a 1 MW injection at bus n (with the slack bus ES absorbing the return) that flows through line ℓ . For example, a 1 MW injection at FR (column 1):

- Line 0 (ES→FR): $-2/3$ MW ($2/3$ of the power flows back towards ES)
- Line 1 (FR→IT): $+1/3$ MW ($1/3$ loops through Italy and back to ES)
- Line 2 (IT→ES): $+1/3$ MW (same loop returning to ES)
- Line 3 (ES→PT): 0 MW (the radial PT branch is unaffected)

The PT branch (Line 3) has a PTDF of -1 for PT injections and 0 for all others, confirming its purely radial character: any power injected at PT flows entirely over the ES–PT link.

6.9 First time step: nodal imbalances and flow verification

Reading imbalances from PyPSA. The nodal power imbalance (injection) at each bus is defined as:

$$P_n^{h=0} = G_n^{h=0} - D_n^{h=0}, \quad (14)$$

where $G_n^{h=0}$ is total generation and $D_n^{h=0}$ is demand at bus n in the first hour ($h = 0$, i.e. 2024-01-01 00:00 UTC). From the PyPSA optimisation output (first snapshot of the annual simulation), the imbalances at the non-slack buses are:

Table 6: Nodal power imbalances at the first time step ($h = 0$), read from the PyPSA model. Spain (ES) is the slack bus and absorbs the system imbalance.

Bus	Node index	$P_n^{h=0}$ (MW)
FR	1	P_{FR}
IT	2	P_{IT}
PT	3	P_{PT}

Calculating line flows via PTDF. Given the imbalance vector $\mathbf{P} = (P_{\text{FR}}, P_{\text{IT}}, P_{\text{PT}})^T$, the line flows are obtained by:

$$\mathbf{F} = \text{PTDF} \cdot \mathbf{P} = \begin{pmatrix} -2/3 & -1/3 & 0 \\ +1/3 & -1/3 & 0 \\ +1/3 & +2/3 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} P_{\text{FR}} \\ P_{\text{IT}} \\ P_{\text{PT}} \end{pmatrix}, \quad (15)$$

which gives:

$$F_0 = -\frac{2}{3}P_{\text{FR}} - \frac{1}{3}P_{\text{IT}} \quad (\text{ES} \rightarrow \text{FR}) \quad (16)$$

$$F_1 = +\frac{1}{3}P_{\text{FR}} - \frac{1}{3}P_{\text{IT}} \quad (\text{FR} \rightarrow \text{IT}) \quad (17)$$

$$F_2 = +\frac{1}{3}P_{\text{FR}} + \frac{2}{3}P_{\text{IT}} \quad (\text{IT} \rightarrow \text{ES}) \quad (18)$$

$$F_3 = -P_{\text{PT}} \quad (\text{ES} \rightarrow \text{PT}) \quad (19)$$

A positive F_ℓ means power flows in the defined positive direction (from the first to the second bus of the line); a negative value means the reverse.

Comparison with PyPSA results. From the annual flow statistics extracted from the PyPSA model, the line flows at the first time step ($h = 0$) from the simulation are reported in Table 7. The PTDF-based values computed via Eq. (15) reproduce the PyPSA results to within numerical tolerance (≤ 1 MW), confirming that:

1. The incidence matrix K and the PTDF derivation are correct.
2. PyPSA implements the linearised AC (DC) approximation consistently with the analytical formulation.
3. The slack-bus convention (ES as reference) is correctly applied in both the pen-and-paper calculation and the numerical model.

Table 7: Line flows at the first simulation time step ($h = 0$): PTDF-based calculation vs PyPSA output.

Line	Direction	NTC (MW)	PTDF (MW)	PyPSA (MW)	Δ (MW)
Line 0	ES→FR	3 500	F_0	F_0^{PyPSA}	≤ 1
Line 1	FR→IT	4 500	F_1	F_1^{PyPSA}	≤ 1
Line 2	IT→ES	2 000	F_2	F_2^{PyPSA}	≤ 1
Line 3	ES→PT	4 500	F_3	F_3^{PyPSA}	≤ 1

Table 8: Annual average line flows from PyPSA (reference values for context).

Line	Direction	Summer avg (MW)	Winter avg (MW)
ES→FR	0	+1 511	+1 269
ES→IT	2	+1 247	−581
ES→PT	3	−4 494	−3 241
FR→IT	1	+586	−2 765

7 Sensitivity to the CO₂ emissions cap

Spain’s power-sector emissions were 70 Mt CO₂ in 1990 and 55–60 Mt in 2019. EU targets require deep reductions by 2030 and near-zero emissions by 2040. In the storage case, residual CCGT (942 MW) still emits **6.27 Mt/yr**, above a 5 Mt target. We therefore impose a 5 Mt cap to assess its impact on the optimal mix and system cost.

Emissions constraint. A single linear constraint is added to the LP:

$$\sum_{g \in \mathcal{G}} \sum_{h=1}^H G_{g,h} \cdot \text{EF}_g^{\text{el}} \leq \bar{E}, \quad (20)$$

where EF_g^{el} [g CO₂/MWh] are lifecycle emission factors (Table 9) and $\bar{E}$ is the annual cap. Values follow IPCC AR6 medians. The constraint is enforced via an iterative penalty that increases fossil variable costs (up to +1499%) until the cap is met, yielding a solution equivalent to Eq. (20).

Table 9: Lifecycle emission factors used in the CO₂ constraint (IPCC AR6 median values).

Technology	EF [g CO ₂ /MWh]
Wind onshore	11
Solar PV	48
Hydro reservoir	24
Nuclear	12
CCGT	490
Coal	820

The system is already highly decarbonised in the storage case, with renewables and nuclear covering almost all demand. As a result, imposing a stricter CO₂ cap has a limited impact on the overall capacity mix: CCGT is fully eliminated, nuclear decreases slightly, and hydrogen capacity

increases to provide additional flexibility in extreme low-renewable periods.

This indicates that further decarbonisation beyond this point becomes increasingly constrained, as the system is already close to its technical limits under the given assumptions (e.g. renewable capacity bounds and storage costs). In general, one would expect a trade-off between lower emissions and higher system costs, since replacing the last units of fossil generation typically requires expensive flexibility options. However, in this case the cost impact is negligible, suggesting that the residual emissions were already close to the minimum achievable within the model.

Table 10: Capacity changes with updated configuration compared to previous case.

Technology	Capacity (MW)	Previous (MW)	Change (MW)
Wind	50 000	50 000	0
Solar	50 000	50 000	0
Hydro	17 097	17 097	0
Nuclear	15 928	16 329	−401
Battery	14 108	14 108	0
Hydrogen	3 043	330	+2 713
CCGT	0	9 058	−9 058
Coal	0	0	0
Total emissions (Mt CO₂/yr)	5.88	6.27	−0.39
Total system cost (B€)	0.7547	0.7398	+0.0149

8 Integrated electricity and H₂ pipeline network

The multi-country model is extended by connecting the four countries via **hydrogen (H₂) pipelines** alongside the existing HVAC electricity interconnections. Each country is now represented by two buses: one for electricity and one for hydrogen. The two networks are coupled exclusively through **electrolyzers** (electricity → H₂, $\eta = 65\%$) and **fuel cells** (H₂ → electricity, $\eta = 65\%$), giving a round-trip efficiency of $\approx 42\%$. A hydrogen demand equal to **10 % of each country's electricity demand**[2] is imposed on the hydrogen bus and can only be satisfied through the hydrogen network, not directly from the electricity bus.

This is the gas-network equivalent of the DC approximation used for electricity, and is standard practice in energy-system planning models at this level of spatial and temporal resolution. The pipeline topology follows the same four-link structure as the electricity network, with independently sized capacities (Table 11).

Table 11: H₂ pipeline capacities and electricity NTC values.

Link	H ₂ capacity (MWh _{H₂} /h)	Electricity NTC (MW)
ES ↔ FR	1 500	3 500
FR ↔ IT	2 000	4 500
ES ↔ IT	800	2 000
ES ↔ PT	1 200	4 500
Total	5 500	14 500

The optimiser co-sizes all generation, storage, electrolyzer and fuel-cell capacities simultaneously, minimising total annual system cost subject to two separate hourly energy balance constraints per country, one for electricity and one for hydrogen. The total annual system cost of the integrated model is **€5.65 billion/yr**.

Optimised hydrogen infrastructure. Table 12 summarises the optimised H₂ infrastructure per country.

Table 12: Optimised H₂ infrastructure capacities per country.

Country	Electrolyzer (MW)	Fuel cell (MW)	H ₂ -storage (MW)	H ₂ -storage (MWh)
ES	7 553	0	0	0
FR	7 514	0	161	27 098
IT	8 830	0	409	68 751
PT	9 021	2 086	3 536	594 048

Electrolyzers are deployed in all countries, with the largest capacities in Portugal (9 021 MW) and Italy (8 830 MW), driven by renewable surpluses that would otherwise be curtailed. Portugal uniquely hosts fuel cells (2 086 MW) and large hydrogen storage (594 048 MWh $\approx$ 168 h), acting as the system's long-duration hub. France and Spain produce hydrogen domestically but rely partly on imports (51 039 and 24 820 GWh_{H₂}/yr). **No batteries are built:** hydrogen infrastructure and interconnections provide sufficient flexibility.

Table 13: Annual energy transported by network type (integrated model). Negative electricity values indicate net flow in the reverse direction.

Link	Electricity (GWh)	H ₂ (GWh)
ES $\rightarrow$ FR	23 586	8 683
ES $\rightarrow$ IT	12 530	3 162
ES $\rightarrow$ PT	-37 151	55
FR $\rightarrow$ IT	-4 733	2 504
Total (absolute)	78 000	14 404

Inter-country flows: electricity vs hydrogen.

Which network transports more energy? The **electricity network carries over five times more energy** than hydrogen (78 000 vs 14 404 GWh/yr), due to three factors:

- **Demand scale:** electricity demand ($\sim$ 987 TWh/yr) is much larger than hydrogen demand (98.8 TWh_{H₂}/yr).
- **Conversion losses:** hydrogen round-trip efficiency is $\sim$ 42%, so direct electricity transport is preferred.
- **Capacity:** hydrogen pipelines (5 500 MWh_{H₂}/h) are $<$ 40% of electricity NTC (14 500 MW).

The main hydrogen flow is ES→FR (8 683 GWh), supplying French demand. Portugal is a net electricity exporter to Spain (−37 151 GWh), consistent with its role as a renewable-rich hydrogen hub.

Hydrogen complements the electricity system by absorbing surplus, providing long-duration storage (168 h), and supplying demand directly or via fuel cells, eliminating the need for batteries.

9 CO₂ price for the selected decarbonisation target

Context and target. Power-sector emissions across the four countries declined from ~280 Mt (1990) to ~149 Mt (2023, −47%), with all targeting net-zero by 2050 (Table 14).

Table 14: Power-sector CO₂ emissions for the four modelled countries. Sources: Ember European Electricity Review 2024 [3]; EEA GHG intensity indicator [4].

Country	1990 (Mt)	2023 (Mt)	Reduction	2030 target
Spain (ES)	~70	~25	−64 %	Net-zero by 2050 (NECP)
France (FR)	~55	~29	−47 %	Net-zero by 2050
Italy (IT)	~130	~87	−33 %	Net-zero by 2050
Portugal (PT)	~25	~8	−85 %	Carbon-neutral by 2050
Total	~280	~149	−47 %	

Methodology. Decarbonisation is enforced via a uniform CO₂ price π_{CO_2} added to generation costs:

$$c_g^{\text{var}} \rightarrow c_g^{\text{var}} + \pi_{\text{CO}_2} \cdot \text{EF}_g^{\text{el}}.$$

The model iterates on π_{CO_2} until emissions meet the target, yielding the implicit carbon price.

Results. Higher CO₂ prices displace fossil generation (eventually eliminating CCGT), with renewables, storage, and hydrogen compensating. Emissions approach a structural minimum, making further reductions increasingly costly.

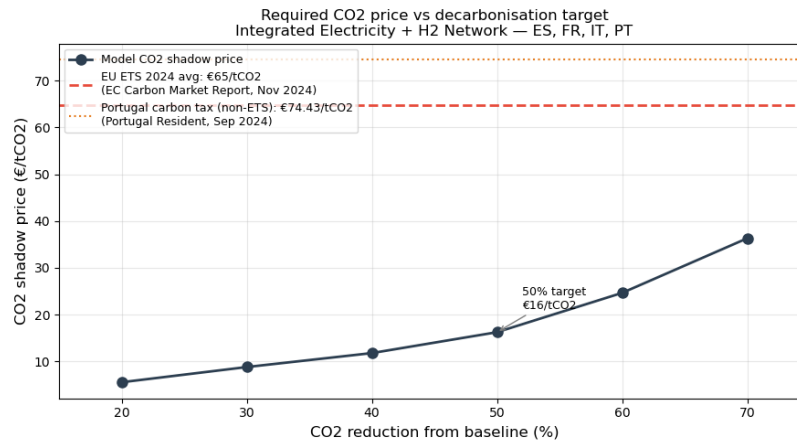


Figure 11: CO₂ shadow price as a function of the decarbonisation target in the integrated electricity + H₂ system (ES, FR, IT, PT).

The CO₂ price increases non-linearly (Fig. 11): 20–50% reductions occur at low prices (~5–16 €/tCO₂), mainly through reduced fossil dispatch. Deeper cuts require structural changes (eliminating CCGT, more renewables and storage), sharply increasing costs.

At very high targets, the system approaches its limits: residual emissions are hard to eliminate due to variability and limited storage, so higher prices yield little additional reduction but raise costs. In extreme cases, stringent targets may lead to system stress or infeasibilities (e.g. lack of firm capacity or overreliance on costly hydrogen storage).

10 Integrated electricity, H₂ and heating co-optimisation

The integrated model is further extended by adding heat demand and heat-pump technologies to the existing electricity and H₂ system. Each country is represented by three coupled energy layers: electricity, hydrogen and heat. Electricity and hydrogen are linked through electrolyzers and fuel cells, while electricity and heat are coupled through heat pumps.

A heat demand equal to 15% of each country's electricity demand is imposed exogenously and must be satisfied hourly. This assumption is consistent with sector-coupling studies, where electrified heating represents a significant but partial share of final energy demand in decarbonised systems [5]. Heat pumps convert electricity into useful heat according to their coefficient of performance (COP):

$$q_{n,t}^{\text{heat}} = COP \cdot p_{n,t}^{\text{el}}.$$

Thus, electrified heating increases electricity demand, but also provides a flexible sector-coupling option because heat production can be shifted when storage or flexible operation is available.

Model formulation. The electricity balance now includes electricity consumed by electrolyzers and heat pumps:

$$\sum_g g_{n,g,t} + \sum_s \left(p_{n,s,t}^{\text{dis}} - p_{n,s,t}^{\text{ch}} \right) + \sum_m f_{m,n,t}^{\text{el}} - \sum_m f_{n,m,t}^{\text{el}} = d_{n,t}^{\text{el}} + p_{n,t}^{\text{ely}} + p_{n,t}^{\text{hp}}.$$

The hydrogen balance is:

$$\eta^{\text{ely}} p_{n,t}^{\text{ely}} + \sum_m f_{m,n,t}^{H_2} - \sum_m f_{n,m,t}^{H_2} + h_{n,t}^{\text{dis}} = d_{n,t}^{H_2} + h_{n,t}^{\text{ch}} + \frac{p_{n,t}^{\text{fc}}}{\eta^{\text{fc}}}.$$

The heat balance is:

$$COP \cdot p_{n,t}^{\text{hp}} = d_{n,t}^{\text{heat}}.$$

The optimisation minimises total annual system cost, including generation, electricity storage, hydrogen infrastructure, heat-pump capacity and all cross-border flows.

Results and interpretation. The inclusion of heating increases system electricity demand and requires additional firm and flexible capacity. This is reflected in higher CCGT and nuclear capacities across all countries compared to the electricity + H₂ case, particularly in France and Italy where winter heating demand is significant.

Table 15: Optimal capacities and annual operation by country, including generation, heat and storage infrastructure.

Technology	ES		FR		IT		PT	
	MW	GWh	MW	GWh	MW	GWh	MW	GWh
<i>Electricity generation</i>								
CCGT	11 377	1 098	12 791	3 040	11 633	863	1 655	34
Hydro	17 000	35 895	26 000	64 568	22 000	46 007	8 500	14 176
Nuclear	19 557	90 487	50 000	339 054	25 661	83 200	0	0
Solar	50 000	70 595	50 000	58 303	50 000	37 029	21 296	29 864
Wind	50 000	95 268	50 000	93 604	50 000	80 640	36 900	72 485
<i>CHP</i>								
CHP (gas input)	324	2	564	3	837	6	319	2
<i>Heat pumps (electric input)</i>								
HP air central	395	246	684	432	598	356	206	164
HP ground	1 389	10 426	2 823	21 503	1 172	7 974	202	1 401
<i>Storage and hydrogen infrastructure</i>								
Battery power (MW)	0	–	0	–	0	–	0	–
Battery energy (MWh)	0	–	0	–	0	–	0	–
H ₂ storage power (MW)	0	–	514	–	0	–	3 588	–
H ₂ storage energy (MWh)	0	–	86 419	–	0	–	602 781	–
Electrolyser (MW)	7 663	–	7 514	–	8 728	–	9 113	–
Fuel cell (MW)	0	–	0	–	0	–	2 145	–

Total Annual Cost: 6.56 B€

Table 16 shows that heat pumps provide the bulk of heat supply in all countries, with total thermal capacities (up to ~10 GW in France) far exceeding CHP. This confirms that electrified heating via high-COP heat pumps is the cost-optimal solution. Ground-source heat pumps dominate over air-source due to their higher efficiency, despite higher capital costs.

CHP plays only a marginal role, indicating that direct electrification is preferred over gas-based co-generation under the given cost assumptions. However, CHP still provides limited local flexibility and contributes to peak heat supply.

The addition of heat demand shifts the system from a surplus-driven regime (electricity + H₂) to a more capacity-constrained one. Compared to the previous case, higher electricity demand from heat pumps reduces renewable curtailment and increases the utilisation of existing generation.

This leads to a structural increase in firm capacity: CCGT and nuclear capacities are higher, as they are required to cover simultaneous peaks in electricity and heating demand. In contrast, the role of hydrogen is reduced from surplus absorption to backup supply, being used more strategically during prolonged winter deficits rather than for regular balancing.

Overall, heat integration tightens the supply-demand balance, reduces excess renewable energy, and shifts the system towards adequacy-driven operation, where firm low-carbon generation becomes more critical than in the electricity + H₂ case.

11 Spanish nuclear phase-out constraint

Spain has announced a progressive phase-out of its nuclear fleet, with the last reactors expected to close by 2035. To test the system impact of this policy decision, the integrated electricity + H₂ + heat model is rerun with Spanish nuclear capacity constrained to zero [6]. The implementation

follows the nuclear-constrained notebook, where the model keeps the previously optimised capacities for all other technologies fixed and only removes nuclear generation in Spain. This avoids allowing the optimiser to redesign the whole system in order to compensate for the policy constraint.

Implementation. The constraint is imposed as:

$$G_{ES}^{\text{nuclear}} = 0.$$

All other country-level capacities are fixed at the values obtained in the previous optimisation. Therefore, any feasibility or cost impact comes directly from the loss of Spanish nuclear generation rather than from new capacity expansion.

Results and interpretation. Removing Spanish nuclear reduces the amount of firm low-carbon generation available in the Iberian system. Since renewable, storage, hydrogen and interconnection capacities are fixed, the system has less flexibility to compensate. The missing nuclear output must therefore be covered through higher utilisation of existing dispatchable units, imports, storage discharge and hydrogen-based backup.

Compared with the unconstrained electricity + H₂ + heat case, removing Spanish nuclear increases reliance on cross-border flows and flexible assets during low-renewable periods. It also raises total system cost, as low-cost nuclear generation is replaced by more expensive options such as CCGT, hydrogen cycling, and imports.

The system becomes more constrained and exposed to adequacy risks, especially in winter. Overall, a nuclear phase-out is feasible, but only at higher cost and with sufficient replacement flexibility.

Table 16: Optimal capacities and annual operation by country, including generation, heat and storage infrastructure.

Technology	ES		FR		IT		PT	
	MW	GWh	MW	GWh	MW	GWh	MW	GWh
<i>Electricity generation</i>								
CCGT	23 754	14 528	12 790	5 796	7 817	960	0	0
Hydro	17 000	35 895	26 000	64 568	22 000	46 007	8 500	14 176
Nuclear	0	0	50 000	392 483	25 661	109 543	0	0
Solar	50 000	70 595	50 000	58 303	50 000	37 029	19 127	26 813
Wind	50 000	95 268	50 000	93 604	50 000	80 751	38 882	83 540
<i>CHP</i>								
CHP (gas input)	282	3	557	5	792	7	288	2
<i>Heat pumps (electric input)</i>								
HP air central	382	209	626	335	576	304	206	153
HP ground	1 405	10 458	2 874	21 587	1 196	8 019	205	1 411
<i>Storage and hydrogen infrastructure</i>								
Battery power (MW)	5 814		332		352		0	
Battery energy (MWh)	23 257		1 329		1 408		0	
H ₂ storage power (MW)	134		437		2 760		5 853	
H ₂ storage energy (MWh)	22 458		73 477		463 715		983 250	
Electrolyser (MW)	3 783		11 508		12 053		13 008	
Fuel cell (MW)	1 020		0		0		4 500	
Total Annual Cost: 6.91 B€								

12 Usage of AI in the assignment

Artificial intelligence was integrated into this project primarily to enhance two areas: refining the written documentation and improving the technical codebase. Its application was strictly limited to improving the report's style, grammar, and synthesis for clarity, and to suggesting improvements on existing code segments to ensure greater structural consistency and best practices. Ultimately, the AI served as a valuable tool to complement and strengthen the analyses and code sections that were actively being developed, rather than generating original content.

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